

Task 1 — 5 Real-World Indicators of Compromise (IoCs)

#	Indicator of Compromise	How it can be detected
1	Suspicious IP address	Firewall/proxy logs may show repeated connections from an IP associated with malicious activity.
2	Unexpected executable file	System logs or EDR logs may show a newly created or executed .exe, .sh, or other suspicious file.
3	Repeated failed logins	Authentication logs can show many failed login attempts from the same source IP in a short period.
4	Unusual outbound traffic	Firewall/DNS/proxy logs may show a compromised machine communicating with an unfamiliar external IP or domain.
5	Unexpected network ports/connections	Network logs can reveal one host attempting connections to many different ports, indicating possible scanning activity.

Task 2 — Splunk: Top 5 IP Addresses

Step 1: Create a sample web log

Create a file named:

- web_access.log

Step 2: Upload the log into Splunk

In Splunk:

- Add Data → Upload → Select web_access.log

For the source type, you can use:

- access_combined

or create/use a suitable web-access source type.

Step 3: Find the top 5 IPs

Run:

- `index=* | stats count by clientip | sort - count | head 5`

If Splunk extracts the client IP as `src_ip` instead:

- `index=* | stats count by src_ip | sort - count | head 5`

Step 4: Create the dashboard

Go to:

- Dashboard → Create New Dashboard

Add a visualization based on:

- `index=* | stats count by clientip | sort - count | head 5`

A bar chart or column chart is a good choice.

Task 3 — Detect Failed Logins Within 10 Minutes

For Splunk, you can use `stats/bin` to group failed authentication events into 10-minute windows.

For example:

- `index=* event_type="login_failed"`
- `| bin _time span=10m`
- `| stats count as failed_attempts by src_ip, _time`
- `| where failed_attempts >= 5`
- `| sort - failed_attempts`

This identifies IP addresses that generated 5 or more failed login attempts within the same 10-minute window.

If your logs use `action=failure` instead:

- `index=* action=failure`
- `| bin _time span=10m`
- `| stats count as failed_attempts by src_ip, _time`
- `| where failed_attempts >= 5`
- `| sort - failed_attempts`

Task 4 — Simulate a Port-Scan IoC

You can safely simulate the log evidence of a port scan without actually performing a disruptive scan.

Create:

- portscan.log

Add:

- 192.168.1.99 07/Sep/2026:09:20:01 port=21
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:02 port=22
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:03 port=23
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:04 port=25
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:05 port=53
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:06 port=80
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:07 port=110
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:08 port=139
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:09 port=443
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:10 port=445
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:11 port=3306
action=connection_attempt
- 192.168.1.99 07/Sep/2026:09:20:12 port=3389
action=connection_attempt

Upload this into Splunk.

Search for the suspicious IP

- `index=* action="connection_attempt"`
- `| stats dc(port) as unique_ports count as attempts by src_ip`
- `| where unique_ports >= 5`
- `| sort - unique_ports`

`dc(port)` means distinct count of ports.

The result might look like:

src_ip	unique_ports	attempts
192.168.1.99	12	12

That is a strong indication of scanning behavior.

Visualize it

Create a dashboard panel using:

- `index=* action="connection_attempt"`
- `| stats dc(port) as unique_ports by src_ip`
- `| sort - unique_ports`

Choose a bar chart.